CURRICULUM VITAE · 2026

郭晨志 KUO, CHEN CHIH

Applying for: ISEP EHS Engineer

b29374665@gmail.com
b29374665-design.github.io/isep
New Taipei · open to shifts, travel, relocation
Military-service exempt · available from 2027

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), with OSH exams scheduled. At the SEED Center I **wrote the TGA / DSC user rules** and run lab safety and equipment management; I also **analysed 975 accident cases** on my own. A chemical engineering background lets me trace hazards back to the process.

Experience

SEED Center, NTUST · MOE Featured Areas Research Center

Lab equipment, asset maintenance & safety officer (during M.S.) — **2025–present**

- **Rules:** wrote the PPE, ventilation and booking rules for the TGA and DSC that every user, including other labs, must follow — e.g. no DSC heating of unknowns without prior TGA decomposition data; volatile solvents only in pressure-rated pans.
- **Process:** changed TGA sample preparation to cut platinum-pan contamination and extend pan life; adopted the vendor's liquid-sample method.
- **Lab safety:** gas-cylinder management and safety walk-throughs, newcomer safety briefings, glovebox O_2 / moisture monitoring and sensor faults; dry-room work under cleanroom gowning procedures.
- **First response:** isolated, handled and cleaned up a broken-glassware cut and a glovebox catalyst-canister leak.
- **Hazardous materials:** lithium metal (water-reactive) in a glovebox, ether / carbonate electrolytes, dismantling faulty Li–$SOCl_2$ cells.
- **Equipment & vendors:** procured and accepted over NT$2M of equipment and service contracts; maintain the TGA Q500 / DSC 2500, usually getting vendor engineers on site within 24 h; ran TGA testing for 6 labs in 3 departments (600+ samples).

Industry project: long-life Li batteries for smart water meters

Project lead — **2025–**

- Wrote the cell-screening SOP and test database; trained the partner's QA / R&D teams to run screening on their own.
- Used EIS, OCV drip and X-ray CT to flag abnormal cells before they become storage or failure risks.

New Taipei Urban Development intern · Tunghai dorm resident assistant

Cross-agency eligibility checks, public inquiries; safety and mediation in a 100-resident dorm — **2021–2024**

Research & Risk Management

- **Accident study:** classified all 1,005 OSHA Taiwan public cases (975 unique) by type: 47% falls; 26 involve chemicals or oxygen deficiency.
- **Hierarchy of controls:** compared hazards before partly replacing AE with DEE; all lithium work in a glovebox; solvents dried and checked by Karl Fischer.
- **Thesis:** partially replacing allyl ether (AE) with diethyl ether (DEE) to study Li-metal interfacial kinetics at −40 °C.
- **Nature Materials**：co-author (TGA measurements), under review · Presenter, 2026 TwIChE Annual Meeting (Dec 5–6, Feng Chia Univ.)

Licences

Wastewater Treatment Specialist (B)
GB040231 — **2026/06**

Indoor Air Quality Specialist
IA020236 — **2026/08**

Waste Disposal Technician (A)
HA160629 — **2026/10**

OSH Management Personnel (B)
Eligibility approved — Exam 2026/11

Occupational Safety Mgmt Specialist (A)
Exam 2027/03

Industrial Hygiene credit programme
Industrial Hygiene 87, Industrial Safety 86; OSH Mgmt & Risk Assessment in progress

Education

M.S. Chem. Eng., NTUST
2025 – 2027 (exp.) · Process Safety & Radiation Protection

B.S. Chem. & Mat. Eng., Tunghai Univ.
2020 – 2024

Skills

EHS fundamentals
GHS labelling · LEL／UEL · Exposure limits · Ventilation · Hierarchy of controls

Thermal hazards & analysis
TGA Q500 · DSC 2500 · EIS

Software
Word／Excel／PowerPoint · OriginPro · Aspen Plus

Next 3–5 years

- Level A Occupational Safety and Hygiene Management Specialist
- English; ISO 14001 / 45001 systems